

Name of the subject	Construction Planning and Management	
L-T-P	3-0-0	
Credits	3	
Name of the Department	Civil Engineering	
Status of the subject	(a) Semester : Autumn (b) Level of subject : 2 nd Year UG (c) Nature of subject : Core (d) Semester to be offered : 3 rd (e) Programme in which the course is included: B.Tech. (H) in Civil Engineering	
Prerequisites	None	
Objectives	The objective of the course is to introduce different facets of construction project management. It is enhance the interest of students by presenting modern tools for making fast, safe and quality construction.	
Names of the faculty members of the department who have the necessary expertise to teach the course	All faculty members of Civil Engineering	
Any overlap with existing subjects	None	
Recommended Text Books	1. Construction Project Management, Theory and Practice, Kumar Neeraj Jha, Pearson, Delhi, 2011 2. Construction 4.0 : An Innovation Platform for the Built Environment, Edited by Anil Sawhney, Mike Riley, Javier Irizarry, Routledge	
Topics to be Covered	Name of the Topic	Hourly Break-up
(1)	Project Organization, Construction Company, Structure of Construction Organization	2
(2)	Construction Economics, Economic Decision-Making, Cash flow diagrams	3
(3)	Client's estimation of project cost, Approximate method of estimation, Types of estimates	3
(4)	Construction Contract, Contract document, Bidding Process	3
(5)	Construction Planning, Critical Path Method	3
(6)	Project Scheduling and Resource Levelling	3
(7)	Contractor's Estimation of Cost and Bidding Strategy	3
(8)	Construction Equipment Management	2
(9)	Construction Accounts Management	3
(10)	Construction Material Management	2
(11)	Construction Quality Management	2

(12)	Construction Safety Management	2
(13)	Overview of Construction 4.0	3
(14)	Digital Ecosystems in the construction industry, BIM	3
(15)	Visual and virtual progress monitoring in Construction 4.0	3
(16)	Construction Claims, Disputes and Project Closure	2